MARCUS ESCALONA

PRODUCT & SOFTWARE ENGINEER · FULL-STACK, C# / .NET, AI/LLM & VOICE AI
Mexico City, MX · Open to hybrid (CDMX) or remote (LATAM / USD) · Native English & Spanish
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PROFESSIONAL SUMMARY

Product and software engineer and technical co-founder with 6+ years across product, systems and infrastructure engineering, who has taken several products from zero to production. I own problems end to end: scoping with non-technical stakeholders, designing the architecture, building in React, TypeScript, Node.js, Python and C# / ASP.NET Core, integrating third-party APIs and shipping. On the backend I design REST APIs, relational data models and business rules (validation, versioning, approval workflows and audit trails) on SQL Server and PostgreSQL, covered by automated tests and CI. As co-founder and CTO of a mobility startup I took the product from zero to early B2B traction; I also built an AI voice and messaging product on LLMs, speech-to-text and telephony/WhatsApp APIs, and founded a real-time 3D sales platform for real-estate developers. I lead engineering for a 200+ user platform suite and work AI-native with Claude Code and Cursor, reviewing every generated change before merging. Native English and Spanish speaker with a background in physics and audio engineering.

TECHNICAL SKILLS

Frontend React · TypeScript · JavaScript (ES6+) · Next.js · Zustand / Redux · Tailwind CSS · Vite · HTML5 · CSS3
C# / .NET C# · .NET 10 · ASP.NET Core Web API · Entity Framework Core (migrations, interceptors, optimistic concurrency) · FluentValidation · API versioning · ProblemDetails
Backend & APIs Node.js · Python microservices · REST API design · Webhooks (signature verification, idempotency) · Supabase (Auth, RLS, Realtime, Edge Functions, pg_cron) · Firebase · Auth & role-based access
Databases SQL Server · PostgreSQL · SQL · Query tuning (execution plans, sargable predicates, filtered and covering indexes) · Relational data modelling · Prisma · Firestore
AI / LLM & Voice Claude API (Haiku / Sonnet) · LLM prompting & evaluation · Vapi voice agents · Speech-to-text (Deepgram / Whisper) · OCR (Tesseract.js)
Integrations Twilio · WhatsApp Cloud API · Resend
Design SOLID · Domain-driven design (aggregates, state machines) · Strategy and Template Method patterns · Dependency injection · Clean architecture
Testing & DevOps xUnit · NSubstitute · FluentAssertions · Testcontainers · Integration testing · Docker · GitHub Actions CI · Vercel · Netlify
3D & Performance Three.js · React Three Fiber · WebGL · Canvas API · Procedural geometry · Custom physics · Render-loop & memory profiling
Workflow Git · GitHub · Pull-request workflows · Agile · Claude Code · Cursor · AI-native development

PROFESSIONAL EXPERIENCE

Lead Product Engineer · Escuela Berta Von Glümer Oct 2023 – Present

- Architected and shipped a suite of five React and TypeScript applications serving 200+ students and staff, defining the frontend architecture, shared component system and deployment pipeline end to end.
- Built CIT-Sync (public demo: github.com/MarcusPicazo/cit-sync), a real-time platform for room and equipment reservations with role-based approval, hardware inventory and software requests (React, Firebase), replacing manual spreadsheet workflows across departments.
- Led delivery of EBVG-Hoot, a real-time multiplayer assessment platform, designing an event-driven data layer on Firestore onSnapshot to keep game state synchronised across concurrent players.
- Owned the 3D engineering track: three interactive learning environments in Three.js and WebGL (a solar system explorer, a colour-mixing art studio and a parkour game with custom collision physics), profiling the render loop and memory use to hold a stable frame rate on constrained school hardware.
- Gathered requirements directly from teachers and directors and translated them into working tools; built Duolingo-style English diagnostic tests as web apps.

Co-founder & CTO · Taylor Mobility Inc. apptaylor.com 2026 · on hold

- Co-founded a safety-first mobility platform for women, students and night-shift workers in Latin America, with a B2B2C model: schools and companies subscribe and offer safe transport to their people as a benefit.
- As CTO, architected the platform end to end: an Android app for riders and drivers (ride tiers, driver matching, SOS, live location sharing, trusted contacts) and a Windows desktop panel where organizations track rides, schedule recurring routes and manage staff.
- Designed the trust-and-safety layer: driver identity and background verification, per-trip facial matching against official ID, encrypted GPS monitoring and a 911 escalation flow when a route deviates.
- Took it to market with my co-founders (CEO and CSO): a 150-person waitlist of drivers and riders, 30 interested local businesses and school commitments for the 2026–2027 cycle, with operations planned for Mexico City, Portland and London.

Systems Engineer · Getting Out, US corrections-technology platform 2022 – 2023
- Built Python automation services, REST APIs and microservices for a US platform that lets families send money to incarcerated people's accounts, streamlining internal data-processing workflows.
- Designed SQL data models and a service-oriented architecture for operational reporting in a payments domain where accuracy and auditability are non-negotiable.

Audio Engineer (Remote) · Black Fox Productions 2019 – 2025
- Delivered broadcast-ready audio post-production for international cinema chains, running asynchronous review cycles and signal routing across Pro Tools, Ableton Live, iZotope RX and Dante.

EARLIER EXPERIENCE

IT Systems Engineer · Cedros College 2021 – 2022
Network security and infrastructure administration across the corporate network: Active Directory, Fortinet, Windows Server, Group Policy.

Systems Engineer · CONACULTA / Secretaría de Cultura 2016 – 2017
Technical architecture support, QA testing and debugging of custom government software: Linux, SQL, Bash.

SELECTED PROJECTS

Socio · Bilingual AI receptionist for US Latino service businesses github.com/MarcusPicazo/SPEC-Socio
- Architected and built the MVP of a bilingual (EN/ES) AI receptionist for Latino-owned contractors in the US: a Vapi voice agent on Twilio numbers answers English calls, and Claude Haiku summarises and classifies each call and sends the owner a Spanish summary over the WhatsApp Cloud API.
- Built the voice-note-to-quote flow: Spanish voice notes are transcribed, turned into itemised English quotes by Claude Sonnet, rendered to PDF in Node.js and delivered by email (Resend) with a public accept link.
- Multi-tenant backend on Supabase (PostgreSQL, RLS, Edge Functions, pg_cron) with verified webhooks, atomic claim-before-process handling against duplicate events and background processing; 111 automated tests passing.

Tezontle 3D · Real-time 3D sales platform for real-estate developers (founder) github.com/MarcusPicazo/aura-demo
- Founded the company and built its product: a reusable engine for interactive 3D unit selectors for pre-sale developments (React Three Fiber, Three.js, Zustand), with towers generated procedurally from JSON polygon configs and units colour-coded by live availability.
- Sales portal and lightweight CRM on Supabase with RLS and Realtime (admin and agent roles, inventory, reservations, prospects, audit history): a unit marked as sold updates on every screen instantly. Engine-first architecture cuts a new client from ~6 weeks to 1–2.
- Built the client-facing demo, landing and website, and set up go-to-market: sales-team recruiting, LinkedIn and WhatsApp prospecting. Repos: aura-demo · aura-landing · tezontle-web.

CIT-Sync Scheduling Rules API · C# / ASP.NET Core github.com/MarcusPicazo/cit-sync-rules-api personal project on the CIT-Sync public demo
- Versioned scheduling-rules service in ASP.NET Core, EF Core and SQL Server: reservations with a state machine, rule sets per school, building or resource with a Draft → Pending approval → Active → Retired lifecycle and a four-eyes rule.
- Automatic audit trail via an EF Core SaveChanges interceptor (before/after values and actor, same transaction); rowversion plus If-Match prevents lost updates. Strategy-pattern rule engine behind POST /rules/evaluate.
- Tuned the overlap query on 100,000 seeded rows: a filtered covering index cut reads from 2,290 to 6–10 pages (3–4 ms warm). Unit and integration tests (xUnit, Testcontainers) run in GitHub Actions.

Bodega Abierta · open-source WMS for food banks (RapiJazz A.C.) github.com/MarcusPicazo/bodega-abierta
Multi-tenant warehouse management system for Mexican food banks (Apache-2.0). ASP.NET Core API on SQL Server with EF Core global query filters for tenant isolation, donation intake into inventory lots, FEFO allocation, pickup slots, versioned policies with approval and audit trail; offline-first PWA with OCR and barcode intake planned. Wrote the product spec and UX for ~35 screens.

NIMIO Levels github.com/MarcusPicazo
Gamified team progression platform with code-based authentication, role-scoped access and an audited action history. React, TypeScript and Supabase; access codes stored hashed, with row-level security enforcing per-member scope.

EDUCATION

Completed B.Eng. Systems Engineering, Universidad de México y España (UEM), diploma in process · B.Sc. Music & Innovation, Universidad Panamericana
Coursework B.Sc. Physics, Universidad Autónoma Metropolitana (through final year) · M.Sc. Astrophysics

CERTIFICATIONS & LANGUAGES

Certifications IBM Full Stack Developer · IBM Machine Learning & Python · Google Digital Marketing · EF SET C2 English
Languages English (native) · Spanish (native) · Italian (B2) · Catalan (B2)